SAFWAN MUHAMMED SHEREEF

Software Engineer | Python & AI Developer

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PROFESSIONAL SUMMARY

AI/ML-focused Software Engineer and CS graduate with hands-on experience across computer vision, generative AI, and backend development through 3 internships and independent research. Currently building production-grade async REST APIs at Freshivores using FastAPI. Passionate about the full ML lifecycle- from training custom models and integrating LLM APIs to deploying real-world AI applications. Seeking an AI/ML or applied AI engineering role.

EDUCATION

B.Tech in Computer Science Engineering **2021 - 2025**
College of Engineering Kidangoor, Kerala

EXPERIENCE

Software Engineer Intern | *Freshivores* **Feb 2026 - Present** | Bengaluru, Karnataka

- Architected and delivered **5-10 asynchronous RESTful API endpoints** using Python and FastAPI, ensuring low latency and reliability for core application features.
- Integrated frontend interfaces with backend services; wrote type-safe code and optimized SQL queries to reduce technical debt and improve database performance.

Python Developer Intern | *Singularis Software Technologies* **Sep 2024 - Apr 2025** | Kochi, Kerala

- Built scalable automation scripts using advanced Python (OOP, multithreading); applied **Linear Regression** for data trend prediction and automated CSV/Excel workflows, **reducing manual effort by 30%**.

AI & Computer Vision Intern | *Techmaghi* **Apr 2023 - May 2023** | Kochi, Kerala

- Designed and deployed real-time object detection networks using **TensorFlow and OpenCV**; fine-tuned preprocessing pipelines, **improving detection accuracy by 20%**.

KEY PROJECTS

Independent Research: Domain-Specific SLM | *Python, PyTorch* *In Progress*

- Designing and training a **Small Language Model from scratch** in PyTorch - implementing custom attention mechanisms and neural architecture without pre-trained base models on an online store dataset.
- Iteratively refining the training pipeline and architecture based on loss analysis (current: 0.324), gaining deep hands-on understanding of data quality requirements, tokenization, and model optimization.

NutriScan AI - Smart Health Companion | *Python, Gemini Vision API, Computer Vision*

- Integrated **Google Gemini Vision API** to classify meal images in real-time, automating nutrition logging and reducing manual food-tracking effort.
- Built an **AI Chef** module that generates personalised recipes from detected ingredients, demonstrating end-to-end generative AI product development.

Fruit & Vegetable Identifier | *TensorFlow, Python, Deep Learning*

- Trained a **CNN from scratch** with custom data preprocessing and augmentation pipelines, achieving **85% classification accuracy** across multiple produce categories.
- Deployed a web interface providing real-time nutritional breakdowns and calorie estimates from live model predictions.

SRA Groups - Attendance & Management System | *FastAPI, React Native, React (Vite), Firebase*

- Independently built a full-stack workforce management system- comprising a **FastAPI backend**, React admin dashboard, and React Native mobile app with **GPS geofencing** for location-validated worker check-ins.
- Implemented role-based access control, **Firebase** integration, audit trail logging, monthly analytics with overtime calculation, and a CRM module to manage brokers, deals, and wage payments.

CFTA – Carbon Footprint Tracking App | *Android (Java), Laravel, Machine Learning*

- Built a full-stack CO_2 monitoring platform with a native Android app, **Laravel admin dashboard**, and ML regression models to forecast emission trends from travel data.

TECHNICAL SKILLS

Languages	Python, JavaScript, Java, SQL
AI / ML	PyTorch, TensorFlow, OpenCV, Gemini Vision API, Scikit-learn, Deep Learning, Computer Vision
Backend	FastAPI, Node.js, Laravel, REST APIs, SQL Query Optimization, Docker, AWS
Frontend	React, React Native, Vite
Tools	Firebase, Git, GitHub Actions, Postman, Linux/Bash
Concepts	OOP, DSA, Generative AI (LLMs), Model Training & Optimization, RBAC, CI/CD

CERTIFICATIONS & ACHIEVEMENTS

- **Google Cloud AI Infrastructure Series**: Certified in AI Hypercomputers, Cloud GPUs/TPUs, Model Deployment, Storage, and GCP Networking.
- **IELTS** Band Score **7.0** | **Cloud Computing** – IEDC GISAT | **Ethical Hacking Associate (EHA)** EC-Council (IIRDC)